Level 4



SIX ON THE RAIL

```
FAB-42 > program loaded: night_shift_FINAL_v3_really_final.fab
FAB-42 > RUN? [y/N] _
```

FAB-42 is finally complete: six emitters, one for each direction, riding on a rail along the border.

Each emitter can move along the border, toggle on and off or just sit there quietly.

The night shift left a stack of programs. No comments, no notes, and no idea what any of them will leave on the board.

Before anyone presses RUN on the real machine, simulate each program and find out.

The nanofabricator now has 6 emitters, one for each direction.

- Emitters always start in their corner
- Emitters can be moved along the border, but not rotated
- Multiple emitters can be in the same place at the same time
- Emitters can be active or inactive
- Active emitters produce laser beams
- Emitters are inactive at the start
- If they are activated, they stay active until they are deactivated

Your task is to simulate a program that controls the emitters.

- A program is a list of instruction lines
- An instruction line has 6 instructions, one for each emitter
- Instructions are explained on the next page



Initial emitter positions

```
F T - T B -
```

Instruction	Description
F	move emitter one cell in clockwise direction
B	move emitter one cell in counterclockwise direction
T	toggle emitter state: activate if it was inactive, deactivate if it was active
-	no operation

Each tick, one instruction line is processed

1. All 6 instructions are executed simultaneously
2. Matter is created and destroyed

- An example is provided on the next page
- You are given a list of radii and programs
- At the start of each program the board is empty
- **Simulate each program**






```
---2-*-*-3---
--*-0-0-0-*--
-*-0-0-0-0-4-
*-0-M-M-0-M-*
-1-0-M-M-0-*-
--6-0-0-0-*--
---*-5-*-*---
```

Example state

Tick N




```
---2-*-*-3---
--*-0-0-0-*--
-*-0-0-0-0-4-
1-0-M-M-0-M-*
-*-0-M-M-0-*-
--6-0-0-0-*--
---*-*-5-*---
```

F T - T B -
is executed




```
---2-*-*-3---
--*-0-0-0-*--
-*-0-M-M-0-4-
1-0-M-0-0-M-*
-*-0-M-M-0-*-
--6-0-0-0-*--
---*-*-5-*---
```

Matter is created
and destroyed

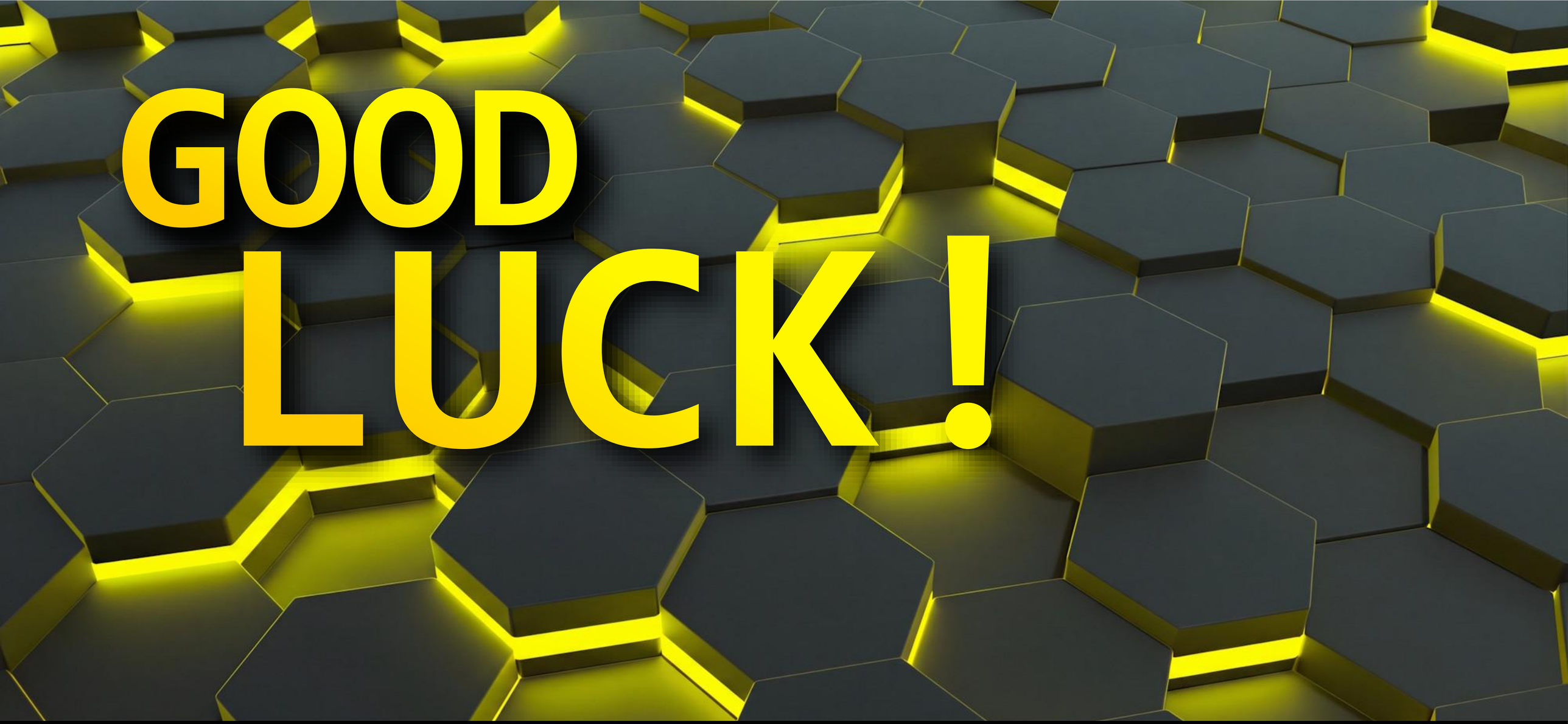
Tick N+1

Input

Name		Description	Example
N		Number of programs	3 2 3 F T T F - - T - - T - - T B - B T - 3 6 B B B B B B - - - T T - - - T - T - - T - T - - T - T - - - ...
repeated N times	R	The radius of the board	
	L	**Number of instruction lines**	
	Program	**A list of L instruction lines** **Each line contains 6 instructions, separated by spaces**	
	Empty line between programs		

Output

Name		Description	Example
repeated N times	Board	The character representation of the board *You can print the border with emitters or without. Both will be judged as correct.*	--*-*-*-- -*-M-M-*- *-M-O-O-* -*-M-M-*- --*-*-*-- ---*-*-*-*--- --*-O-O-O-*-- ...
	Mandatory empty line between boards		



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